# Place Value Power — numbers to 100 000 and beyond

NAME _____ DATE _____ SCORE / EFFORT _____

**Learning intentions — by the end of this worksheet I can:**

- read, write and say numbers to 100 000 (and beyond!)
- explain the value of every digit and rename numbers in different ways
- compare, order and round large numbers
- predict whether an answer will be odd or even without calculating

## Part A · Warm-up ⏱ ~5 min

*Answer as quickly as you can. No working out needed.*

| | | |
|---|---|---|
| a) 7 × 8 = ____ | b) 6 × 9 = ____ | c) 45 + 38 = ____ |
| d) 100 − 37 = ____ | e) double 36 = ____ | f) half of 90 = ____ |
| g) 250 + 375 = ____ | h) 9 × 9 − 1 = ____ | |

## Part B · Learn ⏱ ~8 min

### The place-value chart

| Hundred thousands | Ten thousands | Thousands | Hundreds | Tens | Ones |
|---|---|---|---|---|---|
| | 4 | 7 | 3 | 0 | 6 |

**47 306** is read *forty-seven thousand, three hundred and six*.

- The **digit** 4 has a **value** of 40 000. The digit 7 has a value of 7 000.
- The **0** is a *placeholder* — it shows there are no tens. Without it, the number would be 4 736!
- We leave a small space (not a comma) between groups of three digits: 47 306.

**Each place is 10 times bigger than the place to its right.**
10 ones = 1 ten · 10 tens = 1 hundred · 10 hundreds = 1 thousand · 10 thousands = 1 ten thousand · 10 ten thousands = 1 hundred thousand

### Expanded form

47 306 = 40 000 + 7 000 + 300 + 6

### Renaming (the same number, said differently)

47 306 = 47 thousands and 306 ones = 473 hundreds and 6 ones = 4 730 tens and 6 ones

### Rounding

To round to the nearest **thousand**, look at the **hundreds** digit (the one to its right).
**0–4 → round down. 5–9 → round up.**

> 38 462 → hundreds digit is **4** → round down → **38 000**
> 38 562 → hundreds digit is **5** → round up → **39 000**

### Odd and even

A number is **even** if its ones digit is 0, 2, 4, 6 or 8. Otherwise it is **odd**.

## Part C · Practice ⏱ ~25 min

### C1 ⭐ Fluency

**1. Write the numeral.**
a) forty-seven thousand, three hundred and six _____
b) ninety thousand and ninety _____
c) sixty thousand, six hundred and six _____
d) one hundred and four thousand, five hundred _____

**2. Write in words.**
a) 38 215

b. 70 408

c. 15 050

**3. What is the value of the digit 7 in each number?**
a) 27 315 _____ b) 73 102 _____ c) 50 172 _____ d) 714 000 _____

**4. Expanded form.**
a) 63 507 = _____
b) 40 093 = _____
c) 80 000 + 400 + 6 = _____
d) 9 + 20 000 + 700 = _____ *(careful — the parts are jumbled!)*
e) 5 000 + 30 + 90 000 = _____

### C2 ⭐⭐ Understanding

**5. Rename.**
a) 47 300 = _____ hundreds
b) 47 300 = _____ tens
c) 25 000 = _____ thousands
d) 52 316 = 4 ten thousands + _____ thousands + 316

**6. Order from smallest to largest.**
45 099 · 45 909 · 45 990 · 49 059 · 40 959

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**7. Write <, > or = in each box.**
a) 30 000 + 400 ☐ 34 000
b) 6 thousands + 12 hundreds ☐ 7 200
c) 99 999 ☐ 100 000
d) 15 tens ☐ 1 hundred and 5 tens

**8. Complete the rounding table.**

| Number | Nearest 10 | Nearest 100 | Nearest 1 000 | Nearest 10 000 |
|---|---|---|---|---|
| 38 462 | | | | |
| 71 995 | | | | |
| 45 050 | | | | |

**9. Halfway numbers.** What number is exactly halfway between...
a) 40 000 and 50 000? _____ b) 46 000 and 47 000? _____ c) 38 500 and 39 000? _____




d. Estimate the numbers the arrows point to. A _____ B _____ C _____

**10. Odd or even?** Predict *without* working out the answer.
a) 3 457 + 2 891 _____ b) 12 × 37 _____ c) 135 × 77 _____ d) 999 + 1 000 + 1 001 _____
e) Explain why **odd + odd** is always even. (Hint: think of an odd number as pairs plus one left over.)

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## C3 ⭐⭐⭐ Reasoning

**11. Digit cards.** Use each of these cards **exactly once** to make a 5-digit number.




a. the largest number _____
b. the smallest number (it can't start with 0!) _____
c. the smallest **odd** number _____
d. the number closest to 50 000 _____

e. How many different numbers between 70 000 and 71 000 can you make? List them.

**12. Who am I?**

- I am a 5-digit **even** number.
- My thousands digit is 0.
- My hundreds digit is the largest odd digit.
- My ten thousands digit is double my ones digit.
- My tens digit is my ten thousands digit plus my ones digit.

I am _____. Is there more than one answer? How do you know?

**13. True or false? Explain.**

a) A 5-digit number is always bigger than a 4-digit number.

b. If you round 64 999 to the nearest thousand and to the nearest ten thousand, you get the same answer.

## Part D · Problem Solving ⏱ ~12 min

**14. Footy crowd.** A crowd of **58 437** people watched a football match in Perth.

a) Round the crowd to the nearest thousand: _____ and nearest hundred: _____

b) A newspaper headline said "ALMOST 60 000 FANS!" Is that fair? Explain.

c. How many more people would have been needed to make exactly 60 000? _____

**15. The odometer.** A car's odometer (distance counter) shows **49 996 km**.




a. What will it show after the car drives 7 more km? _____

b. The next time all five digits are the **same** digit, what will the odometer show? _____ How many more km is that from your answer to (a)? _____

WORKING

**16. Three towns.** Town A has 34 782 people, Town B has 34 827 people and Town C has 34 278 people.

a) Order the towns from largest to smallest population. _____

b) How many more people live in the largest town than in the smallest? _____

WORKING

**17. Counting jumps.**

a) Start at 96 450 and count on by 1 000 five times. Where do you land? _____

b) Start at 103 205 and count back by 10 000 three times. Where do you land? _____

## 🔗 Connections — place value meets measurement and money

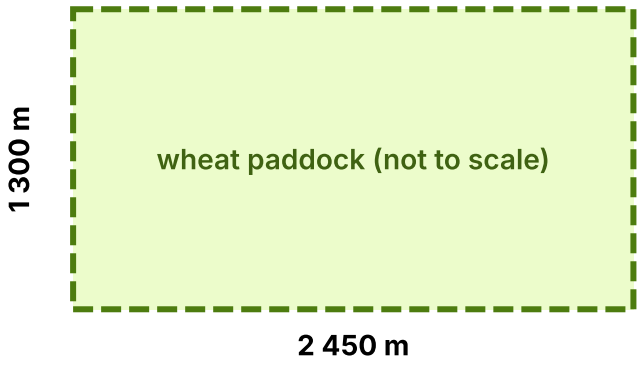
*You already know about metres, centimetres, money and perimeter. Place value works there too!*

**X1. Measurement.** There are 100 cm in 1 m and 1 000 m in 1 km.

a) 4 m 35 cm = _____ cm b) 2 708 cm = _____ m _____ cm c) 12 345 m = _____ km _____ m

**X2. Money.** A school fun run raised **$23 450**. It is paid in $100 notes and $10 notes only, using the **fewest** notes possible. How many of each? _____ × $100 and _____ × $10

**X3. Perimeter + rounding.** A farmer's paddock is a rectangle.



a. Round each side to the nearest **thousand** metres and use the rounded numbers to estimate the distance around the paddock (its perimeter). _____

b. Now round each side to the nearest **hundred** metres and estimate again. _____

WORKING

c. The real perimeter is 7 500 m. Which estimate was closer? Why?

**X4. Odd and even in real life.** 37 students line up in pairs for a bus. Will everyone have a partner? Explain using odd and even.

## Part E · Challenge ⏱ ~10 min

**18. Into the millions.** Write 4 503 060 in words. What is the value of the 5?

**19.** How many 3-digit numbers have digits that add up to 5? (*Example: 203.*) Find them **all** — be systematic!

**20.** Use the digits 1, 2, 3, 4, 5 and 6 once each to make **two 3-digit numbers**.

a) Make their **sum** as large as possible. _____ + _____ = _____

b) Make their **sum** as small as possible. _____ + _____ = _____

c) What did you notice about where the big digits had to go?

## Self-check

| I can... | 😀 | 🙂 | 😕 |
|---|---|---|---|
| read and write numbers to 100 000 | | | |
| explain the value of each digit | | | |
| rename numbers (e.g. 473 hundreds) | | | |
| round to the nearest 10, 100, 1 000 and 10 000 | | | |
| predict odd or even | | | |

## ⚡ Fast-finisher bank ⏱ ~35 min

*Keep going! ⭐⭐⭐⭐ questions are very tricky — even adults find them hard.*

**F1 ⭐⭐ What number is…**
a) 10 000 more than 95 380? _____ b) 1 000 less than 100 450? _____
c) 100 more than 39 950? _____ d) 10 less than 70 005? _____

**F2 ⭐⭐ Estimate** by rounding each number to the nearest thousand first:
38 712 + 21 460 ≈ _____ + _____ = _____

**F3 ⭐⭐** A whole number rounds to 6 000 (to the nearest thousand). What is the smallest it could be? _____ The largest? _____

**F4 ⭐⭐⭐** A whole number rounds to 3 450 (nearest 10) **and** to 3 400 (nearest 100). List every number it could be.

**F5 ⭐⭐⭐** If you write out every whole number from 1 to 100, how many times do you write the digit **7**? _____ Explain how you counted.

**F6 ⭐⭐⭐ Palindromes** read the same forwards and backwards (e.g. 12 321).
a) How many 5-digit palindromes are there between 40 000 and 41 000? _____
b) What is the largest **even** 5-digit palindrome? _____

**F7 ⭐⭐⭐ Counters.** A place-value chart has four columns: Thousands, Hundreds, Tens, Ones. Each counter you place is worth 1 in its column (so a counter in the hundreds column is worth 100).
Using **exactly 3 counters**, list every number you can make. How many are there? How many are odd?

**F8 ⭐⭐⭐⭐** Can you choose **five** numbers from this list (you may use a number more than once) that add up to exactly **30**?
1 3 5 7 9 11 13 15
Explain your answer.

**F9** ⭐⭐⭐⭐ On a number line, A is at 24 000 and B is at 36 000. C is halfway between A and B. D is halfway between A and C. Where is D? _____

WORKING

**F10** ⭐⭐⭐⭐ **Investigation — Egyptian numbers.** About 5 000 years ago, Ancient Egyptians wrote numbers with picture symbols:

| Value | Symbol (description) |
|---|---|
| 1 | a stroke \| |
| 10 | a heel bone ∩ |
| 100 | a coil of rope |
| 1 000 | a lotus flower |
| 10 000 | a finger |
| 100 000 | a tadpole (frog) |
| 1 000 000 | a kneeling god with raised arms |

They repeated a symbol as many times as needed, so 23 was ∩∩|||.

a) How many symbols would an Egyptian need to write **23 145**? _____

b) Which number less than 100 000 needs the **most** symbols? _____

c) Our system uses only ten digits (0–9). Explain why the **position** of a digit and the digit **0** make our system so much quicker.